Curriculum Vitae

Hong-Yi Zhang

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Education & Experience

• 2018.08-present	PhD in physics, Rice University	Advisor: Mustafa Amin
• 2014.09-2018.06	BS in Physics, Nanjing University	

Summer school:

• 2022.06-2022.07	TASI Lectures, CU Boulder
• 2020.08	SLACK Summer Institute, Stanford (online)

Research experience:

• 2018.10-present	Research assistant, Rice University	Research advisor: Mustafa Amin
• 2017.08-2017.10	Summer research, HKUST	Research advisor: Yi Wang

Research interests

I am interested in cosmology, astrophysics, high-energy physics and computational physics. More specifically, I wish to understand the nature of dark matter, the formation of large-scale structure, nonlinear dynamics of solitons and chaos, compact objects as a probe of new physics, effective field theory, the thermal history of the early universe, quantum mechanics in curved spacetime, and modified and quantum gravity.

My previous and current research involves the effective theory for scalar and vector fields, the theory and phenomenology of axion-like and dark photon dark matter, nonlinear dynamics of nontopological solitons, and numerical simulations with General Relativity.

Honors & Awards

• 2022	William & Elva Gordon Fellowship	Rice University
	For great academic and research achievements in astrophysics.	
• 2022	Graduate Student Travel Award	Rice University
• 2020	Chuoke Award	Rice University
	For great performance in coursework and speedy progress in research.	

Before PhD:

• 2018	Outstanding Graduates	Nanjing University
• 2017, 2016, 2015	National People's Scholarship	Nanjing University
• 2017	Scholarship of Excellence	Institute of High Energy Physics, CAS
• 2017	Successful Participant	Interdisciplinary Contest in Modeling
• 2016	National Endeavor Fellowship	Nanjing University
• 2015	First Prize for Combat	Nunchakus Competition in Nanjing
• 2015	First Prize in Shunde District	Contest for Integrated Curriculum

• 2014-2022	Guohua Scholarship	Guohua Memorial Middle School

Before undergraduate:

• 2013	Third Prize	China National Physics Olympiad
• 2013	First Prize in Guangdong Province	China National Mathematics Olympiad
• 2013	Second Prize	Guangdong Biology Olympiad
• 2012	First Prize in Scientific Research	Beijiao Innovation Competition
• 2008	First Prize in Gansu Province	"Youth China" Piano Competition

Service

• Referee for Physical Review Letters.

Talks

• 2022.06	A singularity problem for interacting massive vectors TASI Lectures, CU Boulder
• 2021.08	Gravitational effects on oscillon lifetimes COSMO'21, UIUC
• 2021.04	Gravitational effects on oscillon lifetimes TSAPS Conference, Texas A&M-Corpus Christi
• 2020.11	Classical decay rates of oscillons TSAPS Conference, UT Arlington

At Rice University:

• 2022.11	Nonminimally coupled dark photon dark matter
• 2022.04	Dark photon substructure and formation of vector solitons
• 2021.10	Vector oscillons
• 2021.04	Beyond Schrodinger-Poisson: Nonrelativistic effective field theory for scalar dark matter
• 2020.04	Classical decay rates of oscillons

Fun talks:

• 2022.06	SQUAAAAAK TASI Lectures, CU Boulder

Teaching

• 2021.01-2021.05	TA for "Galaxies and cosmology". Lecturer: Mustafa Amin.
• 2020.08-2020.12	TA for "Statistical and thermal physics". Lecturer: Randy Hulet.
• 2020.01-2020.05	TA for "Computational physics". Lecturer: Frank Toffoletto.
• 2019.08-2019.12	TA for "Quantum field theory". Lecturer: Mustafa Amin. I also gave a lecture on the Lorentz group representation for quantum fields.
• 2019.01-2019.05	TA for "Modern physics". Lecturer: Randy Hulet.

Mentoring:

• 2021.10-2022.08	Undergraduate student: Hang Su (University of New Haven)

List of publications

See my INSPIRE for more information.

1. Z.-G. Mou and **H.-Y. Zhang**,
   A singularity problem for interacting massive vectors,
   Phys.Rev.Lett. 129 (2022) 15, 151101, 2204.11324.
2. **H.-Y. Zhang**, M. Jain and M. A. Amin,
   Polarized vector oscillons,
   Phys.Rev.D 105 (2022) 9, 096037, 2111.08700.
3. B. Salehian, **H.-Y. Zhang**, M. A. Amin, D. I. Kaiser and M. H. Namjoo,
   Beyond Schrödinger-Poisson: Nonrelativistic effective field theory for scalar dark matter,
   JHEP 09 (2021) 050, 2104.10128.
4. **H.-Y. Zhang**,
   Gravitational effects on oscillon lifetimes,
   JCAP 03 (2021) 102, 2011.11720.
5. **H.-Y. Zhang**, M. A. Amin, E. J. Copeland, P. M. Saffin and K. D. Lozanov,
   Classical decay rates of oscillons,
   JCAP 07 (2020) 055, 2004.01202.